Corrigendum for "Base Rate Neglect and the Diagnosis of Partisan Gerrymandering"

Sanford C. Gordon
New York University

Sidak Yntiso
University of Rochester

February 10, 2026

The set of constraints on feasible parameter values used in the R script to generate Figure 5 (p. 206) was incomplete, producing an inaccurate depiction of the regions of feasible (ν, σ) pairs that can generate $DA(\lambda) = 0$. The expressions in the accompanying text contain no errors that we are aware of.

The complete set of constraints is:

1. For any ν, σ must lie between $\max\{0, 2\nu - 1\}$ and $\min\{1, 2\nu\}$.
2. For feasible values of ν and σ (as determined by 1 above):
   (a) If $\nu \leq 0.5$ and $\sigma \geq \nu$, then $\theta^R - \theta^D \in \left[\frac{\upsilon-\sigma}{1-\nu}, 1 - \frac{\sigma}{2\nu(1-\nu)}\right]$;
   (b) If $\nu \leq 0.5$ and $\sigma < \nu$, then $\theta^R - \theta^D \in \left[1 - \frac{\sigma}{\nu}, 1 - \frac{\sigma}{2\nu(1-\nu)}\right]$;
   (c) If $\nu > 0.5$ and $\sigma \geq \nu$, then $\theta^R - \theta^D \in \left[\frac{1-\sigma}{2\nu(1-\nu)} - 1, \frac{\nu-\sigma}{1-\nu}\right]$; and
   (d) If $\nu > 0.5$ and $\sigma < \nu$, then $\theta^R - \theta^D \in \left[\frac{1-\sigma}{2\nu(1-\nu)} - 1, 1 - \frac{\sigma}{\nu}\right]$.

Substituting the constraints on $\theta^R - \theta^D$ into equation (7) in the original text and solving for σ, we find that the feasible region lies for $DA(\lambda) = 0$ lies between $\sigma = \nu$ and

$$\sigma = \begin{cases} \max\{0, \frac{(4\lambda\nu - 3\lambda - 2\nu + 2)\nu}{1-\lambda}\} \text{ if } \nu \leq 0.5 \\ \min\{1, \frac{1 - 4\lambda\nu^2 + 5\lambda\nu + 2\nu^2 - 2\lambda - 2\nu}{1-\lambda}\} \text{ if } \nu > 0.5. \end{cases} \tag{1}$$

Implementing the changes described above, we offer a revised version of Figure 5 that better captures how the set of values consistent with $DA(\lambda) = 0$ varies with λ. Specifically, while a DA standard moderately up-weighting packed votes (λ=2/3), as suggested by Tapp (2019) and Nagle (2017), appears to permit roughly symmetric deviations from proportionality as one moderately up-weighting cracked votes (λ=1/3), the two additional cases reveal stark dissimilarities in the range of permissible seat-vote shares between standards that greatly privilege packed (λ= 5/6) or cracked votes (λ=1/6). Note the correspondence between this figure and Figure 2 in Nagle (2017). We thank Professor Nagle for spotting the issue.

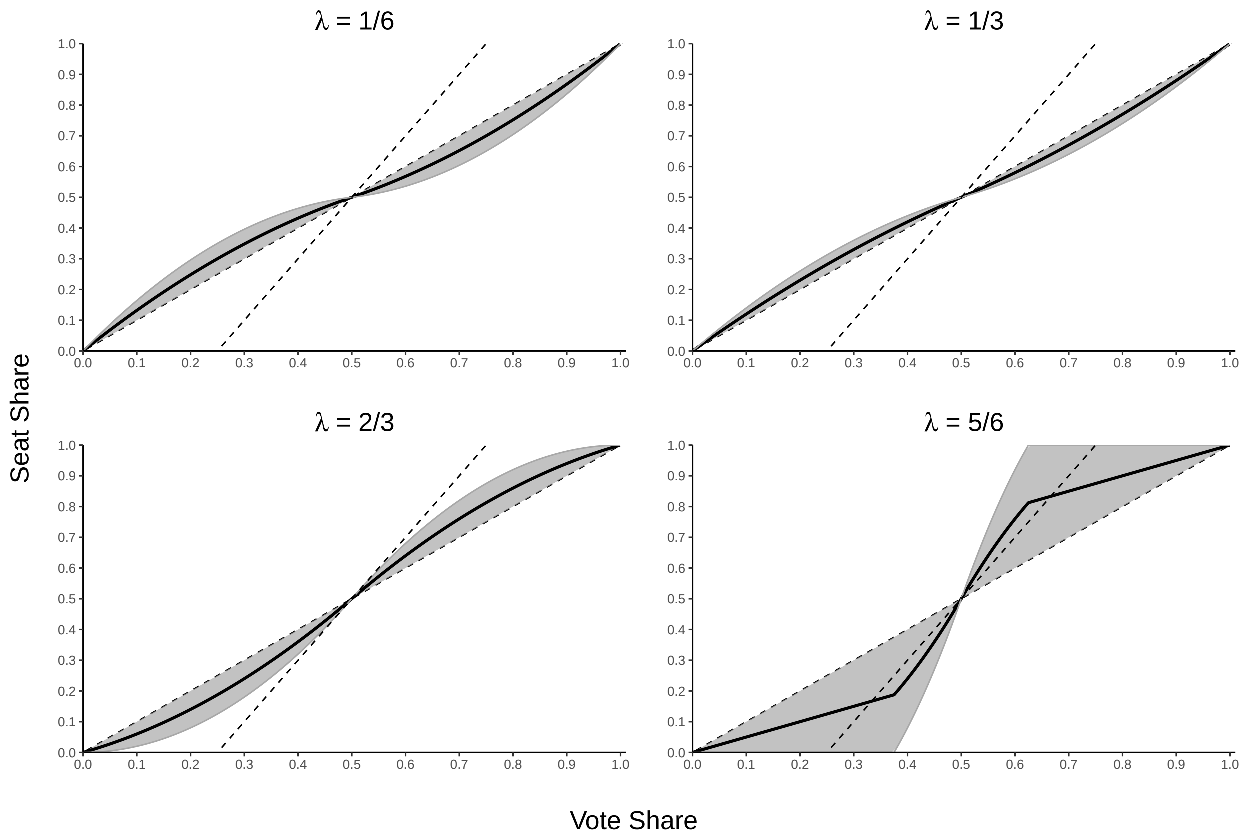

Figure 5: Bounds on Dilution Asymmetry-Minimizing Seatshares. Note: Gray regions represent seatshares that achieve $DA^P(\lambda) = 0$ given voteshare and some feasible (θ^D, θ^R) pair. The thick black curve denotes the seatshare that achieves $DA^P(\lambda) = 0$ for an intermediate feasible (θ^D, θ^R) pair. Dashed lines denote proportionality and double proportionality. See text for discussion.

References

Nagle, John F. 2017. "How Competitive Should a Fair Single Member Districting Plan Be?" *Election Law Journal* 16 (1): 196–209.

Tapp, Kristopher. 2019. "Measuring Political Gerrymandering." *The American Mathematical Monthly* 126 (7): 593–609.